Predicting Coral Bleaching events using climatic factors along with ENSO cycles

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The candidate confirms that the work submitted is his/her own and that appropriate credit has been given where reference has been made to the work of others.

Chapter 4

Results

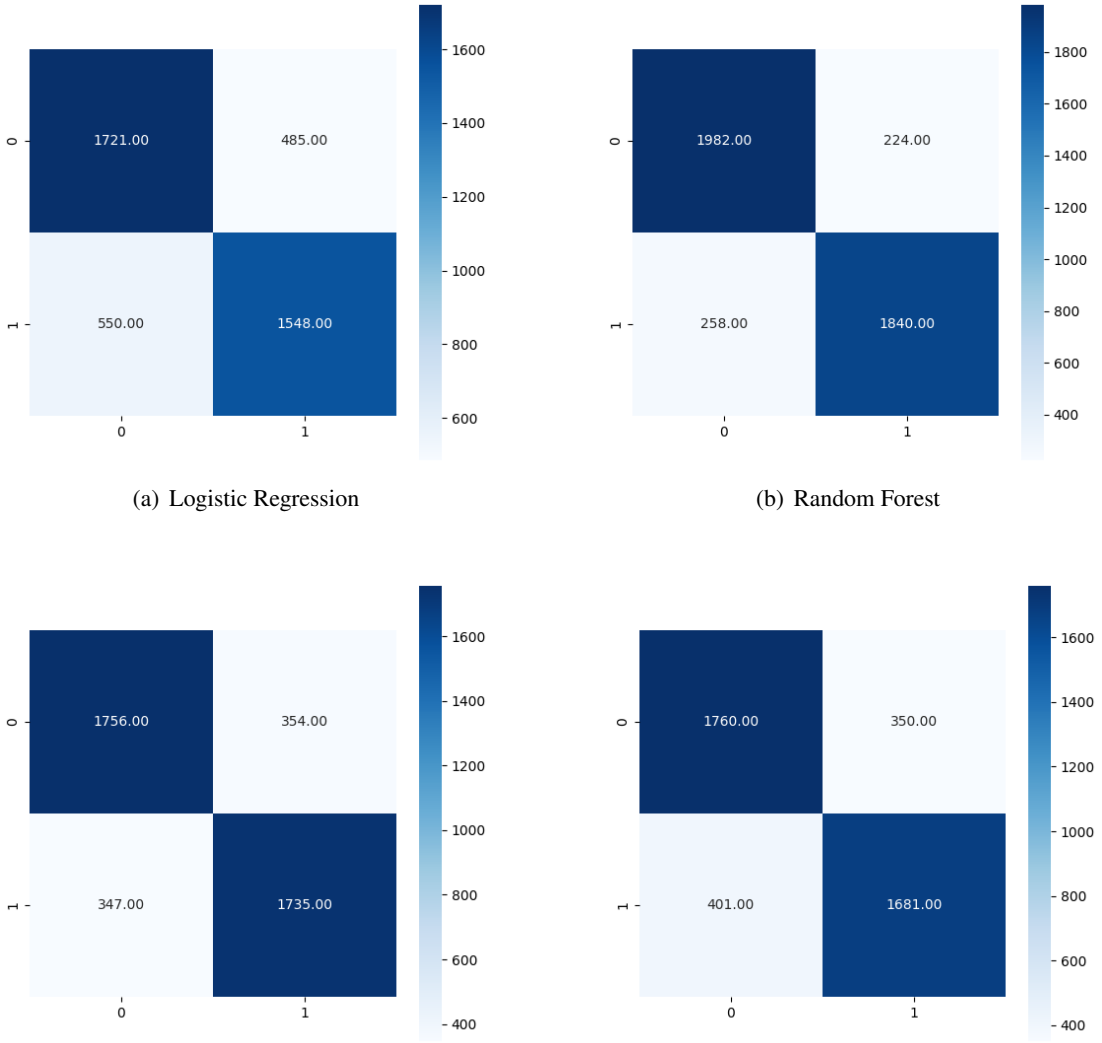
Let us look at the results of the classification model first. So, as mentioned above four classification models were run: Logistic Regression, Random Forest, Decision Tree and K nearest neighbours. Among these models the random forest classifier performed the best with the accuracy of 88.38 percent on the test data set. The aim of this classifier was to classify bleached corals instances as 0 and non-bleached coral instances as 1. Additionally, to identify the most important features that influenced the performance of the model the feature importances method was implemented which is a part of the random forest module in python. To evaluate the performance of classification model there are various known techniques such as confusion matrix, accuracy, ROC curve, precision, recall and F1-score. Our models will be checked on all of these metrics to find the model whose performance is above average in all or most of the aforementioned evaluation metrics. In this process, we check how well the models are able to predict the correct class labels which is 0 and 1 in our case (where 0 is not bleached and 1 is bleached).

4.1 Confusion Matrix

A confusion matrix is a table that categorises predictions according to whether they match the actual labels. It has four components: True Positive, True Negative, False Positive and False Negative. The class of interest is the positive class, while all others are negative class.

- True Positive(TP): Instances that are correctly classified as the positive class.
- True Negative(TN): Instances that are correctly classified as the negative class.
- False Positive(FP): Instances that are incorrectly classified as the positive class.
- False Negative(FN): Instances that are incorrectly classified as the negative class [15, p. 92].

From the confusion matrices shown in figure 4.1 for logistic regression, random forest, decision trees and K nearest neighbours. It can be seen that random forests have a better ability



(a) Logistic Regression

(b) Random Forest

(c) Decision Trees

(d) K Nearest Neighbours

Figure 4.1: Confusion matrices of classification models.

to capture and distinguish between the classes than logistic regression. An ideal classifier would have no false positives and false negatives. But since it is not possible to achieve that in real life, we go with the one which has fewer instances of false positives and false negatives which in this case is random forest with 224 and 258 instances of false positives and false negatives respectively. The the number of true positives and true negative instances are also more in random forests than in logistic regression.

4.1.1 Interpreting the model performance

From the table we can observe the values of train and test accuracies, precision, recall and F1-score. These values give an idea about the performance of the classification models numerically.

Model Name	Train Accuracy	Test Accuracy	Precision	Recall	F1-score
Logistic Regression	75.25	75.95	0.76	0.73	0.74
Random Forest	99.83	88.80	0.89	0.87	0.88
Decision Tree	99.84	84.97	0.83	0.85	0.84
K Nearest Neighbors	85.42	79.11	0.80	0.75	0.77

Table 4.1: Evaluation metrics for classification models.

The train and test accuracies represent the performance of the model on train and test data set respectively. It is important to assess the performance of the model on test set since that would be an accurate measure of how well the model does on previously unseen data. To interpret F1-score it is important to understand the following key terms:

1. Sensitivity: It is also called as Recall or True Positive Rate. This values tells how many of real positives were correctly predicted [15, p. 94].

$$\text{Sensitivity (Recall)} = \frac{TP}{TP + FN} \tag{4.1}$$

2. Specificity: It is also called as the True Negative Rate. This value tells how many real negatives were correctly classified [15, p. 93].

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{4.2}$$

3. Precision: It aims to answer the question: "How many positive predictions are truly positive?". The difference between precision and recall is that precision focuses on prediction results such as how useful the prediction results are whereas recall focuses on samples such as how complete the predicted results are taking into account all true samples [15, p. 93].

$$\text{Precision} = \frac{TP}{TP + FP} \tag{4.3}$$

4. F1-score: It is a combined measure to assess a given model in terms of both precision and recall together [15, p. 95].

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4.4}$$

Ideally we would want precision and recall values to be one since only then there would be zero false negatives and false positives. But in reality, a good classifier would have the respective values close to one. In this case, the random forest classifier, outperforms other models in every evaluation metric with test accuracy of 88.80 percent, precision of 0.89, recall of 0.87 and f1-score of 0.88.

4.2 Interpreting the ROC curves

A Receiver Operating Curve (ROC) is a diagnostic curve which assesses the overall performance of any given model. It does so by plotting the true positive rate(y-axis) against the false positive rate(x-axis). True Positive Rate(TPR), also known as sensitivity, is the proportion of instances which are actually bleached and are classified correctly. The False Positive Rate also known as specificity, is the proportion of instances that are not bleached but are incorrectly classified as bleached. An ideal classification model, would have FPR of 0 and TPR of 1. The Area Under the Curve (AUC) represents the ability of the classifier to distinguish between the classes which means, higher the AUC the better the model's performance [21]. From the ROC curves shown in figure 4.2, it is evident that the random forest model has performed the best with an AUC value of 0.95.

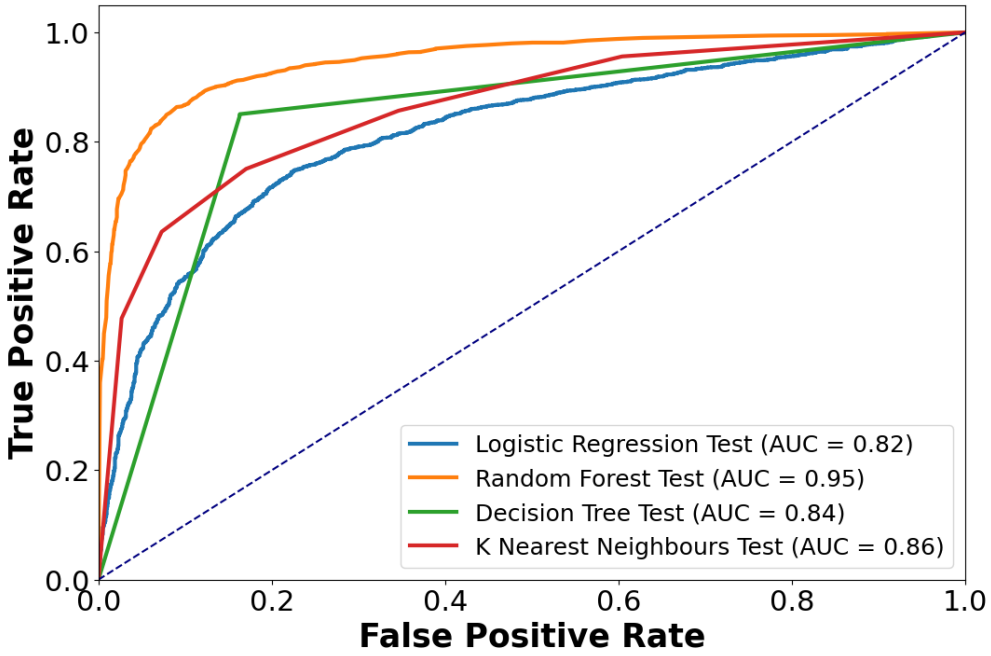


Figure 4.2: ROC curves for classification models.

4.3 Extracting feature importances

The most important feature of the decision tree and random forest classifier is that they provide access to a module called feature importances. It assigns a score to all input features based on how useful they are at predicting the target variable [15, p. 200]. For this particular dataset, feature importance values would be very useful since it has a high dimensionality. As discussed earlier, one of the reasons to build classification model was to determine the most important features. Selecting only the most important features and then training the model will improve

the efficiency of the predictive models. As we can see from the plot of the feature importances in figure 4.3, we are able to identify which features are least relevant to the model. The feature selection was done based on the threashold value of 0.02, which means the features with a score of 0.02 or more were chosen to further check the performance of the best classification model and also in the regression models.

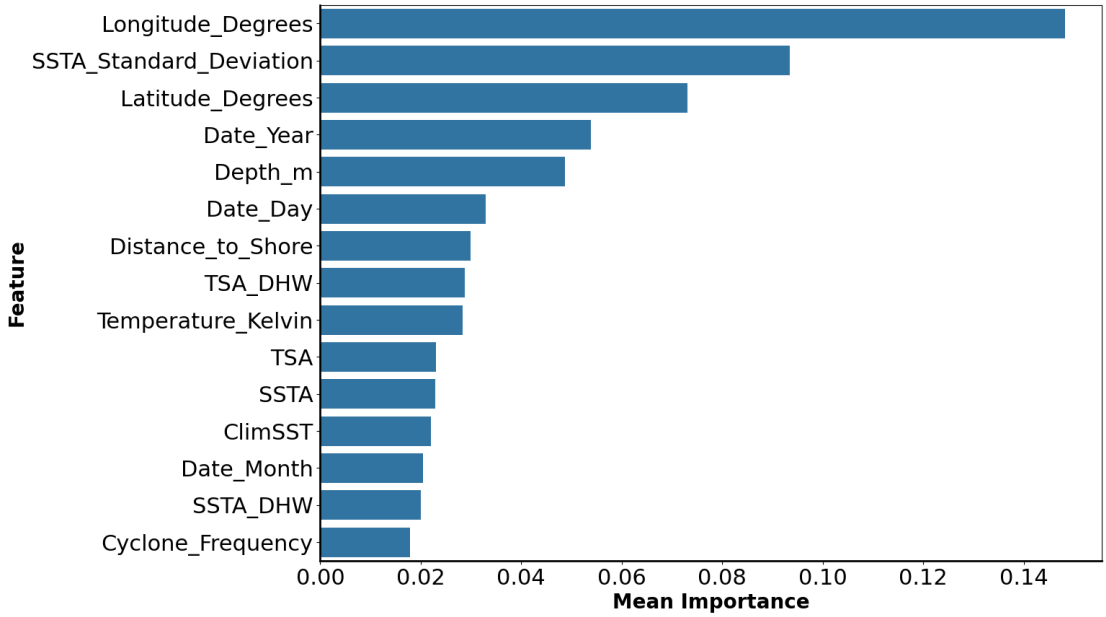


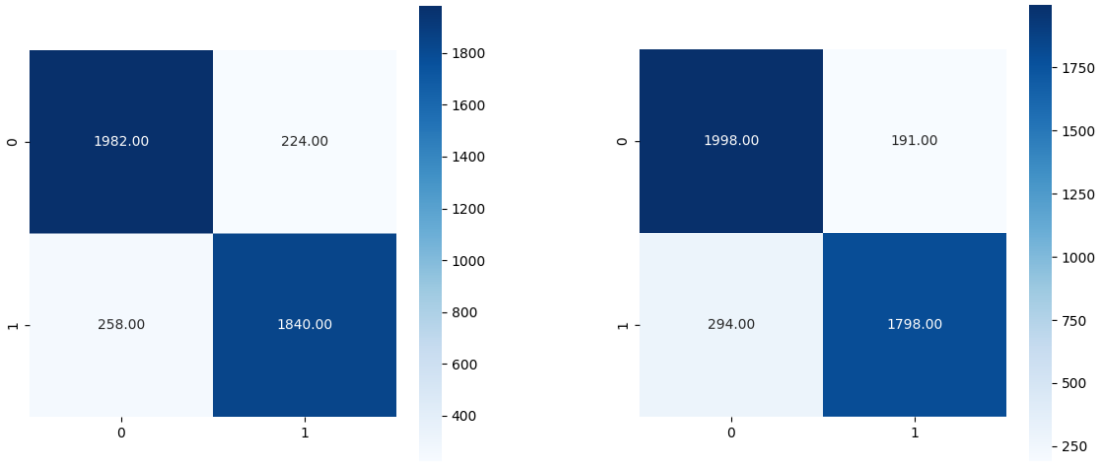
Figure 4.3: Feature importances plot using random forest.

To avoid repetitive features and multicollinearity, a few features were not used. The final features used are: $Longitude_Degrees$, $SSTA_Standard_Deviation$, $Latitude_Degrees$, $Depth_m$, $Date_Year$, $Distance_to_Shore$, TSA_DHW, $ClimSST$,$Percent_Bleaching$.

The feature Date_Day was dropped since there was already the Date_Year feature which has a higher importance score while also specifying the year of each observation. Additionally, $ClimSST$ along with other temperature variables which were in degree Celsius. Hence, to maintain uniformity among features it was opted to disregard $Temperature_Kelvin$ because it represents the value of $ClimSST$ which is the sea surface temperature in Kelvin.

4.3.1 Performance of random forest classifier after selecting the most relevant features

After selection of the most relevant features, our best performing classification model which is Random Forest was re-run and checked if there were any improvements in its ability to capture whether an observation belongs to the not bleached class or the bleached class. The use of confusion matrices to compare the performance provides an easy to interpret method of assessment where the amount of correct and incorrect classifications can be checked for both models.



(a) Before selecting important features

(b) After selecting important features

Figure 4.4: Confusion Matrix of Random forest classifier.

As can be seen from the confusion matrices above, the darker squares representing true values that were correctly classified have slight changes in value. Similarly, there does not seem to be a significant improvement in the negative classes (values that were classified in the wrong group). This result shows that using a selected set of features does not directly influence the performance of our classification model. On the other hand, this can also mean that the set of chosen features represent the predictive power of the model using all the features. This tells us that the set of chosen instances can be used ahead with the reason of logic, relevance in the context of coral bleaching, and results depicting comparable performances with or without the entire set of features.

4.4 Performance of regression models

Now that we are aware of the most relevant features that are responsible for bleaching. We can attempt to identify the exact percentage of bleaching of any given site. These features were used to train three regression models: Linear Regression, Random Forest Regressor and XGBoost Regressor and its performance was evaluated using metrics like Mean Squared Error(MSE), Root Mean Squared Error (RMSE) and R-squared value. The error term represents how the observed value is different from the actual value. Regression algorithms plot the best fit line that minimizes the error term. But not all points lie exactly on the line. Hence, the distance between each data point and the best fit line makes up the error term. Often residuals and error term are used synonmously, there is a very slight difference between the two[13].Residuals demonstrate how the predicted values differ from the actual values while the error term tells how far is the predicted value from the actual value.

- Mean Squared Error (MSE): The MSE value is calculated as the average of squared differ-

ence between predicted value and actual value of the target variable. This value is always non-negative; the closer the value is to zero, the better the performance of the regression model. This is because squaring penalises models for larger error values. For our regression models the MSE value was calculated using the mean_squared_error module from the scikit-learn library which applies the principle of equation 4.5.

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{4.5}$$

Where:

- n is the number of instances
- y_i is the actual value
- $\hat{y}_i$ is the predicted value.

- Root Mean Squared Error (RMSE): It is the square root of MSE value. It estimates the standard deviation of residuals. It is calculated using equation 4.6.

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{4.6}$$

- R-squared value: R-squared is a measure that rewards you for including too many independent variables, and it does not provide any incentive to stop adding more variables. Its value increases every time an independent variable is added to the model. Hence, when a model contains a lot of variables, it tends to capture every noise or random points in the data. This can lead to overfitting which is the model failing to generalise on the whole dataset. This issue is overcome by the adjusted R-squared value which increases only if a new variable improves the model fit [14].

4.4.1 Regression model

Initially, the regression models produced poor results. The linear regression model was the weakest model in our case. This could be due to the fact that even though there are relationships between the target variable and independent variables, it is too complex for the linear regression model to understand. Hence, it was opted to re-run the data on Random Forest regressor and XGBoost Regressor. Even though both these models yielded a very good accuracy on the training data, they failed to generalise the predictions on the unseen test data set which means that they were overfitting. To reduce this overfitting, on both these models hyperparameter tuning was performed using RandomizedGridSearchCV to correctly identify which of the models is the best predictor of the target variable without any overfitting.

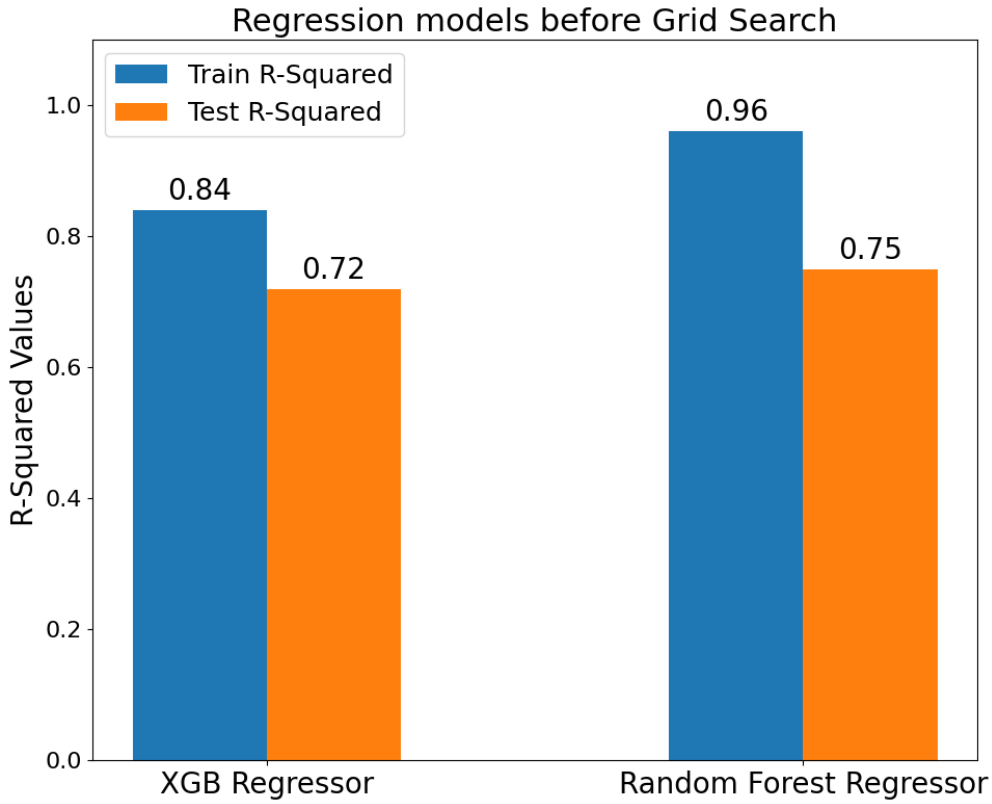


Figure 4.5: R-Squared values of regression models before hyperparameter tuning using Grid Search.

Hyperparameter	Overfit model	generalised model
bootstrap	False	False
max_depth	18	10
max_features	log2	log2
min_samples_split	4	2
min_samples_leaf	2	1

Table 4.2: Hyperparameters for Random Forest Regressor

The hyperparameters used in random forest regressor were learning_rate, max_depth,

max_features, min_samples_split, and min_samples_leaf. max_depth denotes the maximum depth of the tree which is the path between root node (starting node) and leaf node (last node with no further split). max_features is the number of features to take into account when finding the best split. min_samples_split is the minimum number of samples needed to split an internal node. min_samples_leaf is the minimum number of samples required to be at a leaf node. If bootstrap samples are used to build trees it is true, when it is false the whole dataset is used to build each tree.

The hyperparameters used in XGBoost regressor were learning_rate, max_depth,

min_child_weight, colsample_bytree, and gamma. Here max_depth means the same as the max_depth of random forest regressor. min_child_weight is the minimum sum of instance weight

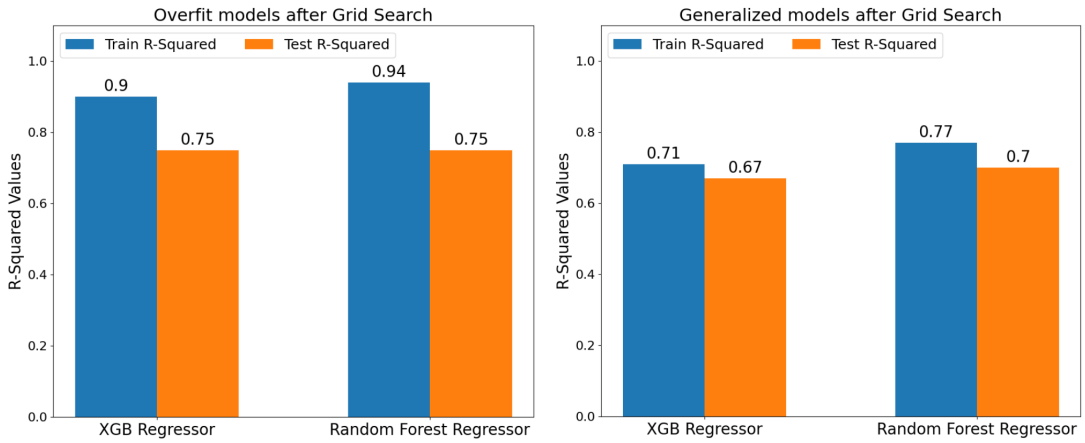


Figure 4.6: R-Squared values of regression models after hyperparameter tuning using Grid Search.

Hyperparameter	Overfit model	generalised model
learning_rate	0.05	0.05
max_depth	12	6
min_child_weight	5	3
gamma	0.0	0.0
colsample_bytree	0.5	0.5

Table 4.3: Hyperparameters for XGBoost Regressor

needed in a child. The lesser this value, lesser the overfitting. colsample_bytree is the subsample ratio of columns when constructing each tree. Minimum loss reduction necessary to make another partition on a leaf node of the tree is denoted by gamma. The learning_rate controls how each new tree contributes to the ensemble prediction. Lower the learning rate, the lesser is the model prone to overfitting.

Before performing GridSearchCV, there was a huge gap between the regressor model's performance on the train dataset and test dataset as shown in figure 4.5. Even though both random forest and XGBoost performed well on the train set, they failed to generalise on the test set, leading to a poor performance on the test set. This indicates that both the regression models are overfitting. Hence, to find which model is the best between these two we need to carry out GridSearchCV to simultaeneously reduce overfitting and choose the best hyperparameters. However, even after performing GridSearchCV on random forest and XGBoost regressors, there was little effect on the R-sqaured values on the test set as indicated in figure 4.6. The reason behind this was a higher value of the hyperparameter max_depth. On reducing the value for max_depth, it was observed that the model could generalise well on unseen data. But this was achieved by compromising on the performance of the model on the train dataset. Ideally, we should always prefer a model that generalises well on unseen data. In this way, the model will not be sensitive to the variance among the input parameters. In this case, the XGBoost regressor

performs better than random forest regressor in generalising on unseen data with a very small margin.